

Answer Key — Physics Grade 10

Sub-Strand 1.1: Pressure • Lesson 2: What is Pressure? Force and Area on Flour

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	A	A	C	C	D	D

USE AFTER: PREDICT PHASE — Predict: brick placed on the large, medium and small face on the tray of flour

Q1. A 2 kg brick (weight 20 N) is placed gently on flour, first on its large face and then on its small face. Which statement about the force the brick presses on the flour with is correct?

- A. The force is greater on the small face because it is narrower.
- B. The force is 20 N on both faces because the weight does not change.**
- C. The force is greater on the large face because it is bigger.
- D. The force on the small face is zero because the brick is balanced.

Correct answer: B. The force is the brick's weight, $W = mg = 2 \times 10 = 20$ N, however the brick is placed. Only the contact area changes.

USE AFTER: OBSERVE PHASE — Observe: converting each face area from cm² to m² in the data table

Q2. The large face of a brick measures 20 cm by 10 cm, so its area is 200 cm². Given that 1 m² = 10,000 cm², what is this area in m²?

- A. 0.02 m²**
- B. 0.002 m²
- C. 0.2 m²
- D. 2 m²

Correct answer: A. Divide by 10,000 to change cm² to m²: $200 / 10,000 = 0.02$ m².

USE AFTER: OBSERVE PHASE — Observe: calculating $P = F/A$ for each face of the brick

Q3. A 2 kg brick ($g = 10$ N/kg) rests on its small face, which has an area of 0.005 m². What pressure does it exert on the flour?

- A. 4,000 Pa**
- B. 1,000 Pa
- C. 0.1 Pa
- D. 0.00025 Pa

Correct answer: A. $F = mg = 2 \times 10 = 20$ N. $P = F/A = 20 / 0.005 = 4,000$ Pa.

USE AFTER: EXPLAIN PHASE — Explain: claim-evidence-reasoning statement on why the small face sinks deepest

Q4. The same brick of weight 20 N makes the deepest dent in flour when it rests on its small face. What is the best explanation?

- A. The small face makes the brick heavier.
- B. The small face pushes with a bigger force than the large face.
- C. The same 20 N is spread over a smaller area, so the pressure is greater.**
- D. Force and pressure are the same thing, so the dent depends only on the force.

Correct answer: C. The weight is 20 N on every face, but a smaller area gives a larger force per unit area, so the pressure is greater and the dent is deeper.

USE AFTER: EXPLAIN PHASE — Explain: quick calculation check (30 N on 0.5 m²)

Q5. A force of 30 N acts at right angles on a surface of area 0.5 m². What is the pressure on the surface?

- A. 0.017 Pa
- B. 30 Pa
- C. 60 Pa**
- D. 15 Pa

Correct answer: C. $P = F/A = 30 / 0.5 = 60$ Pa.

USE AFTER: EXPLAIN PHASE — Explain: writing the definition of pressure as $P = F/A$ with the unit Pa

Q6. Which statement correctly defines pressure and its unit?

- A. Pressure is the force multiplied by the area, measured in pascals.
- B. Pressure is the total force on a surface, measured in newtons.
- C. Pressure is the mass of an object, measured in kilograms.
- D. Pressure is the force acting at right angles per unit area, measured in pascals where 1 Pa = 1 N/m².**

Correct answer: D. Pressure is force acting normally per unit area, $P = F/A$, and its unit is the pascal (1 Pa = 1 N/m²). It is not the same as force.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Lesson 1 knife model with 'same force, small area, $P = F/A$ '

Q7. Mama Wanjiru pushes with the same 40 N force on a sharp knife and on a blunt knife. The sharp knife cuts a tomato easily. Using what you learnt from the brick on flour, what is the best explanation?

- A. The pressure is the same for both knives because the force is the same.
- B. The sharp knife has a bigger force on the tomato than the blunt knife.
- C. The sharp knife has a larger contact area, so it cuts better.
- D. The sharp edge has a much smaller contact area, so the same force gives a greater pressure.**

Correct answer: D. The force is the same, but the sharp edge touches the food over a much smaller area, so $P = F/A$ is greater. This is the same idea as the brick's small face.